J.18

NAME ☞

SCHOOL ☞

TEACHER ☞

NAME	
SCHOOL	
TEACHER	

Pre-Junior Certificate Examination, 2015

Mathematics

Paper 1

Higher Level

Time: 2 hours, 30 minutes

300 marks

For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5		15	
6		16	
7			
8			
9			
10		Total	

School stamp

Running total	

Grade

Instructions

There are 16 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times, you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

(a) **(i)** What is an irrational number?

(ii) π is one of the most identifiable irrational numbers.

The fraction $\frac{22}{7}$ is commonly used to approximate the value of π.

To how many decimal places is this approximation correct?

(b) **(i)** The columns in the table below represent the following sets of numbers:
Natural Numbers ($\mathbb{N}$), Integer Numbers ($\mathbb{Z}$), Rational Numbers ($\mathbb{Q}$),
Irrational Numbers ($\mathbb{R}\setminus\mathbb{Q}$) and Real Numbers ($\mathbb{R}$).

Complete the table by evaluating each of the expressions using $a = 3$ and indicating with a tick (✓) to show to which set(s) of numbers it belongs.
(The first expression has been completed for you.)

Expression	**Evaluated**	**Number Sets**				
		$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}\setminus\mathbb{Q}$	$\mathbb{R}$
a	3	✓	✓	✓		✓
$a^2 - a$						
$\sqrt{a}$						
$a - a^2$						
$\frac{a-1}{a^2}$						
$\frac{\sqrt{a}}{5}$						

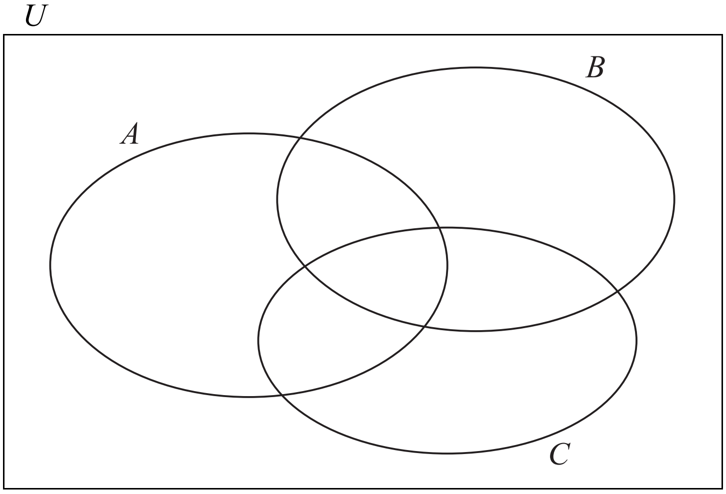
(ii) Place each of the numbers evaluated in part **(i)** in ascending order.

page	running

Question 2 **(Suggested maximum time: 10 minutes)**

$U = \{1, 2, 3, \ldots, 15\}$. A is the set of prime numbers less than 15, B is the set of odd numbers less than 15 and C is the set of factors of 15.

(a) Complete the Venn diagram.



(b) List the elements of each of the following sets:

(i) $A \setminus (B \cup C)$

$A \setminus (B \cup C) =$

(ii) $(A \cup B \cup C)'$, where $(A \cup B \cup C)'$ is the complement of the set $A \cup B \cup C$.

$(A \cup B \cup C)' =$

(iii) Find $\#(A \cup B \cup C)'$.

$\#(A \cup B \cup C)' =$

(c) **(i)** Ellen says that for all sets $(A \cup B) \cup C = A \cup (B \cup C)$.
Is she correct? Give a reason for your answer.

Answer:

Reason:

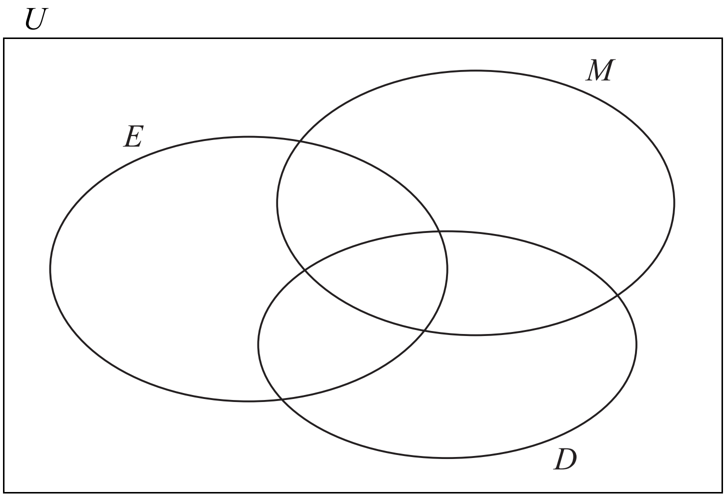
(ii) Name another set in the Venn diagram above which displays the same property.

Question 3 **(Suggested maximum time: 10 minutes)**

150 students went on a school tour to a ski resort in Switzerland. The resort offers three grades of mountain slopes: Easy (E) for beginners, Moderate (M) for more advanced skiers and Difficult (D) for experienced skiers. On their return, the students were asked on which slopes they had skied.

70 students said they had skied on the easy slopes.
60 students said they had skied on the moderate slopes.
55 students said they had skied on the difficult slopes.
10 students said they had skied on the easy and the moderate slopes.
15 students said they had skied on the moderate and the difficult slopes.
12 students said they had skied on the easy and the difficult slopes.
All students said they had skied.

(a) Represent the above information on the Venn diagram.



(b) How many students skied on all three grades of mountain slopes?

(c) How many students skied on only one grade of mountain slope?

(d) Is it reasonable to say that students used this school tour to improve their skiing? Justify your answer.

Answer:

Justification:

page	running

Question 4 **(Suggested maximum time: 5 minutes)**

(a) Factorise fully $3x^2 - 27y^2$.

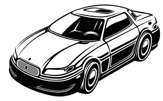
(b) Factorise fully $6pr - 2qs - 3qr + 4ps$.

(c) Use factors to simplify the following:

$$\frac{x^2 + 3x}{x^2 + x - 6}.$$

Question 5 **(Suggested maximum time: 10 minutes)**

A company has a policy of renewing the company cars it supplies to its employees every two years. The company depreciates the value of the cars at a compound rate of 12·5% per annum in its accounts.



(a) Mark receives a new company car which cost €40,000. How much will the car be worth at the end of two years?

(b) At the end of the second year, Mark may choose to keep the car for an additional year and after that he may purchase it from the company for two-thirds of its then value. If Mark takes this option, how much will it cost the company in the third year?

page	running

Question 6 **(Suggested maximum time: 10 minutes)**

(a) Solve for x:

$$5x^2 - 7x - 10 = 0,$$

giving your answers correct to two decimal places.

(b) Hence, or otherwise, solve $5(t^2 - 2)^2 - 7(t^2 - 2) - 10 = 0$.
Give your answers correct to two decimal places.

Question 7 **(Suggested maximum time: 10 minutes)**

(a) In the celebrity TV programme *Splash!*, Keith Duffy completed a dive from a platform, 30 m high. The speed of his dive was timed at 54 km/h.
Calculate the time taken for Keith to complete the dive.

(b) Olympic diver, Tom Daly, repeated the same dive.
It took him $1\frac{2}{3}$ seconds from the instant he left the platform to hit the water.
Calculate the average speed of Tom's dive in kilometres per hour.

(c) Find the ratio of the average speeds of the two divers, giving your answer in its simplest form.

page	running

Question 8 **(Suggested maximum time: 10 minutes)**

The Doyle family of two adults and four children plans to visit the Titanic Belfast museum. The total cost of admission for the whole family is £60. On the same day, the Ryan family of five adults and two children will pay £92 to visit the museum.

(a) Find the cost of admission for an adult and the cost of admission for a child to the museum.

(b) Verify your answers.

(c) The museum offers a family rate of £39 for two adults and two children.
How much would each family save if they avail of this special family rate?

(d) Emma Doyle availed of this special family rate and pays the admission fees for her family by credit card. Later, she notices the amount included on her credit card bill for the museum visit was €67·21. Assuming no commission was charged on the transaction, calculate the exchange rate on that day from sterling (£) to euro (€), correct to the nearest cent.

Question 9 **(Suggested maximum time: 10 minutes)**

(a) Solve the following inequality and show the solution on the number line.

$$-14 < 2 - 4x < 10, \ \ x \in \mathbb{Z}.$$

(b) Simplify $(3x^2 - 7)(5x - 2)$.

(c) Divide $4x^3 - 4x^2 - 7x - 20$ by $2x - 5$.

page	running

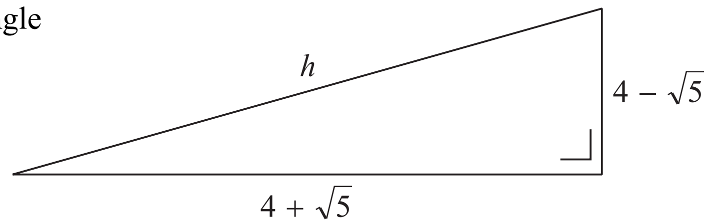
Question 10 **(Suggested maximum time: 5 minutes)**

(a) A Pythagorean triple is a set of three positive integers that can form a right-angled triangle.

(i) Show that the three numbers 20, 21 and 29 form a Pythagorean triple.

(ii) List **two** other examples of Pythagorean triples.

(b) The diagram shows a right-angled triangle which has a base length of $4 + \sqrt{5}$ and a perpendicular height of $4 - \sqrt{5}$.

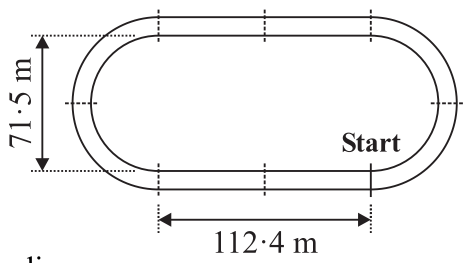


Calculate the value of h, giving your answer in surd form.

Question 11 **(Suggested maximum time: 5 minutes)**

Mo Farah is the current Olympic and World champion at both the 5 000 m and 10 000 m distance events.

He has accepted an invitation to run at a local athletics meeting on the track shown in the diagram which does not conform to international competition regulations.



How many laps of the track will the 10 000 m race need to be?
Show the two approximate positions of the end of the race on the diagram.

Question 12 **(Suggested maximum time: 5 minutes)**

Niamh is a young 100 m sprinter. During training and competition, her speed is regularly monitored and analysed using the formula:

$$d = \frac{(u + v)t}{2},$$

where d is distance, u is initial speed, v is end speed and t is the time interval.

(a) When Niamh practises speed testing, her speed is recorded at five-second time intervals. During one time interval, Niamh's initial speed was clocked at 4 m/s and her end speed was clocked at 7 m/s.
What distance did she run in that time interval?

(b) Write v in terms of d, u and t.

(c) Given that the 100 m event starts from rest, find the minimum end speed at which Niamh must cross the line to finish a race in under 12·5 seconds.

page	running

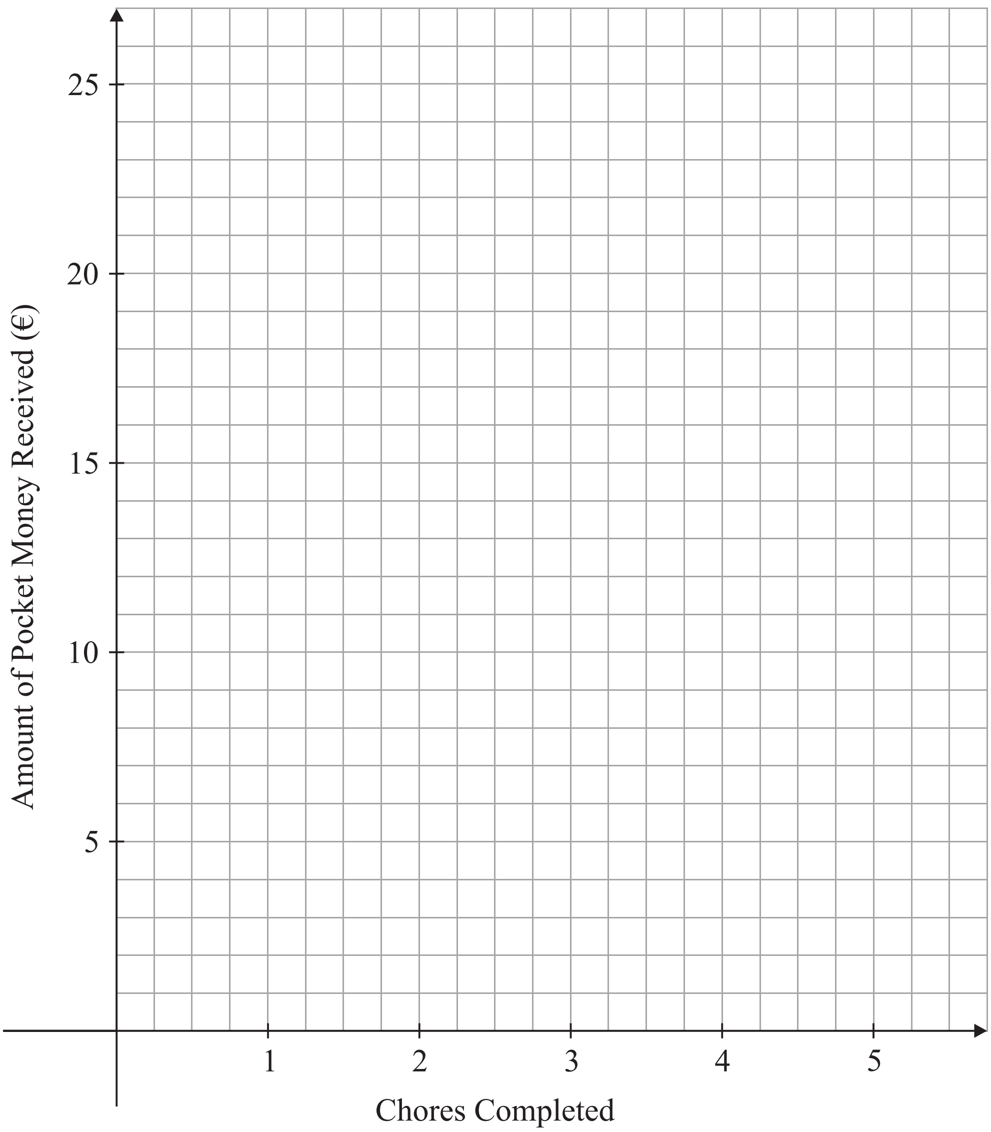
Question 13 **(Suggested maximum time: 15 minutes)**

Adam and Laura are brother and sister who receive pocket money each week.
Adam receives €10 per week and Laura receives €15 per week.

The amount of pocket money each receives increases depending on the number of chores completed at home during that week, as shown in the table below.

Chores completed	Adam (€)	Laura (€)
0	10	15
1	13	17
2	16	19
3		
4		
5		

(a) Complete the table above and draw a graph to show the relationships between the number of chores completed and the amount of pocket money each person receives.



(b) Use your graph to find the number of chores each person must complete in that week in order that both receive the same amount of pocket money.

Adam: ____________________ Laura: ____________________

(c) Write down a formula to represent the amount of pocket money that Adam receives. State clearly the meaning of any letters you use in your formula.

(d) Write down a formula to represent the amount of pocket money that Laura receives. State clearly the meaning of any letters you use in your formula.

(e) Use your formulae from parts **(c)** and **(d)** to verify the answer that you gave to part **(b)** above.

(f) Which person do you think fares better? Give a reason for your answer.

Person:

Reason:

page	running

Question 14 **(Suggested maximum time: 10 minutes)**

(a) The table below shows the growth, in cm, of a plant over a 4-week period.

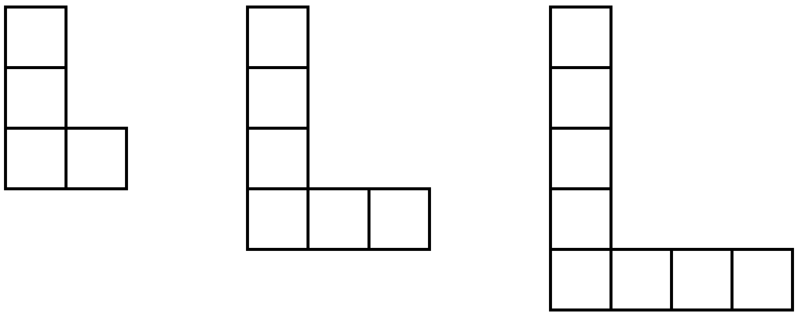
Week	1	2	3	4
Growth (cm)	3	10	21	36

Is the pattern of growth shown in the table linear, quadratic or exponential?
Explain your answer.

Answer:

Reason:

(b) The first three stages of a pattern are shown below.
Each stage of the pattern is made up of square tiles.



(i) Draw the next two stages of the pattern.

(ii) How many tiles are there in Stage 10 of the pattern?

(iii) Find a general formula for the number of tiles in Stage n of the pattern, where $x \in \mathbb{N}$.

(iv) In which stage does the pattern consist of 250 tiles?

Question 15 **(Suggested maximum time: 5 minutes)**

Let g be the function $g : x \mapsto x^2 - 3$, where $x \in \mathbb{R}$.

(a) Find the value of $g(-2)$.

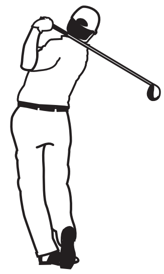
(b) Express $g(2t - 1)$ in terms of t.

(c) Hence, find the values of t for which $g(2t - 1) = g(-2)$.

page	running

Question 16 **(Suggested maximum time: 15 minutes)**

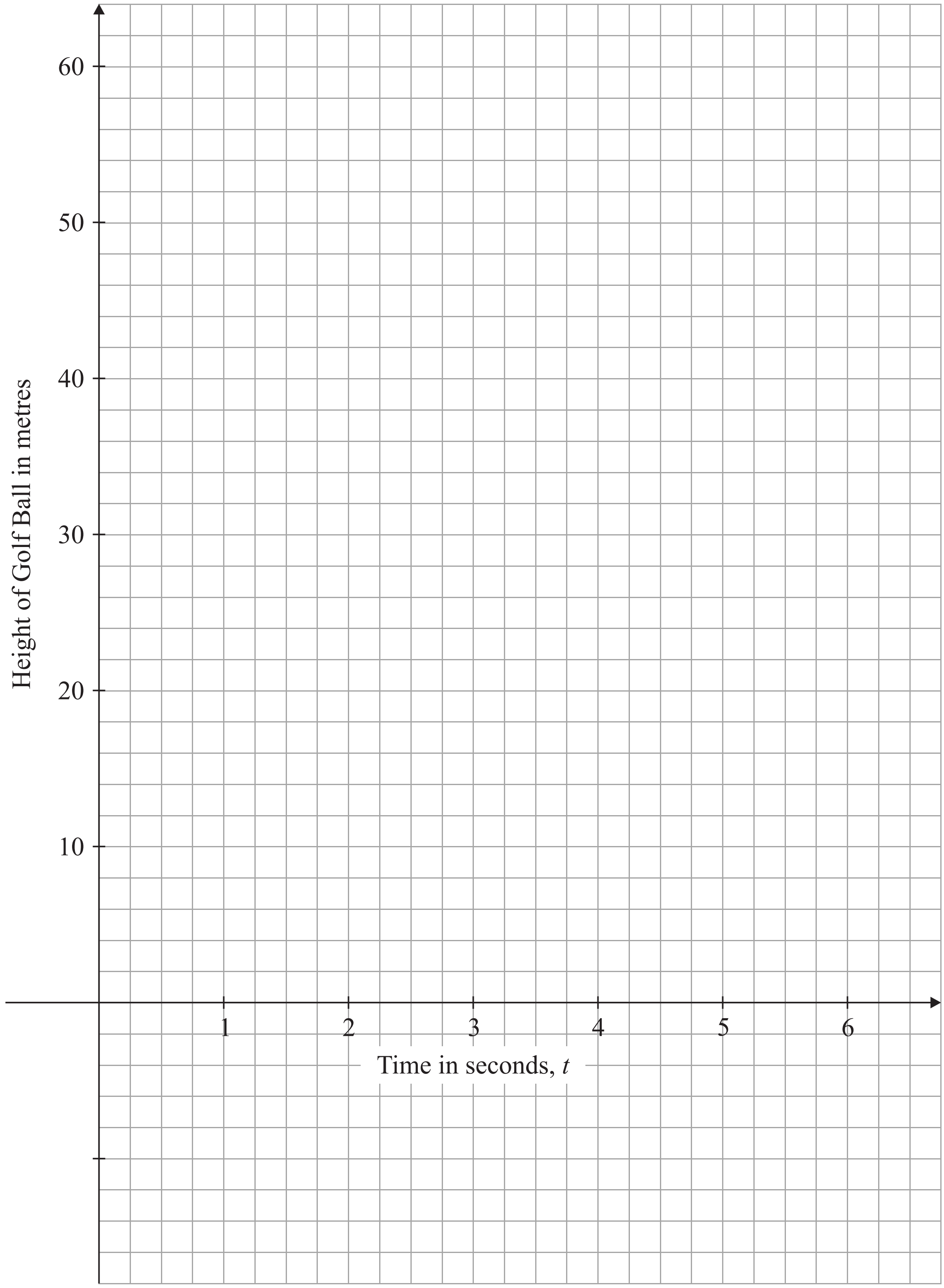
Irish golfer Rory McIlroy is currently the highest ranked player in the world. A group of students have analysed his swing technique. They have come up with a function to predict the height of his golf ball above ground level, after he hits his tee shot.



The height, in metres, of the ball after t seconds is given by $h : t \mapsto 36t - 6t^2$, in the domain $0 \leq t \leq 6$, where $t \in \mathbb{R}$.

(a) On the grid below, draw the graph of $y = h(t)$ in the domain $0 \leq t \leq 6$.

There is room for working out on the next page.



For parts **(b)**, **(c)** and **(d)**, you **must** show your working out on the diagram on the previous page.

(b) Use your graph to estimate the height of the ball after 1·75 seconds. How long will it take before the golf ball reaches this height again?

(c) Imagine Rory took the same tee shot again from an elevated position 16 m above ground level. By extending your graph, estimate the time it would take for the golf ball to hit the ground.

(d) Write down a formula to represent the height of the golf ball above ground level after t seconds for this shot. State clearly the meaning of any letters you use in your formula.

page	running

Pre-Junior Certificate, 2015 – Higher Level

Mathematics – Paper 1

Time: 2 hours, 30 minutes